

Time Travel for Beginners
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1. What, exactly, is Time Travel?

You are travelling through time right now. But that's not how we normally think of time travel, or how we see it portrayed in the movies. So what is the distinctive mark of time travel?

One famous answer from the philosopher David Lewis is that time travel is a numerical discrepancy between **personal time** and **external time**.

Personal time is time “for the time traveler.” Think of it as time on the time traveler's watch.

External time is time “for everyone else.” Think of it as time for the people the time traveler left behind in order to time travel.

Example: someone travels back in time ten years, and it takes thirty years in the time machine to get back in time ten years.

Problem #1: This definition of time travel does not cover all cases, and covers some cases that it should not.

The Lewisian personal time/ external time discrepancy can be viewed as encompassing two “mismatches”:

- (i) a mismatch between the *speed* of the time traveler's bodily processes and the processes in the external timeline
- (ii) a mismatch between the apparent *direction* of the time traveler's bodily processes and the rest of the timeline.

Not all cases of time travel involve both types of mismatch.

Case A: No quantitative discrepancy between personal time and external time.

Straightforward Travel: Tamika pushes the button on her time machine, which will take her backwards in time one hour. The time machine travels at a rate of one hour in personal time per one hour of external time.

Case B: Mismatch in speed of time traveler's bodily processes, but no mismatch in direction.

Slow Travel: Mykaela pushes the button on her time machine, which will take her forward in time one year. The time machine travels at a rate of one hour in personal time per one year of external time.

Case C: Neither a mismatch in speed nor a mismatch in direction.

Reverse Aging: 21-year-old DeShawn presses the button on his time machine, and takes twenty years in personal time to travel back twenty years in external time. However, the time machine ages him backwards at a rate of one year per year, such that his body and bodily processes match those of a one-year-old baby upon his arrival in the past.

2. Paradoxes of Time Travel

"Second time around" time travel is time travel that involves changing the past after it is already set. For example, suppose that you missed your brother's ninth birthday the first time around, and then you travel back in time to attend it the second time around. Then you were both not at the birthday party, and at the birthday party.

The famous *Grandfather Paradox* is a particularly problematic instance of "second time around" time travel: Tim, a time traveler to the past, intends to kill his young grandfather before he can conceive Tim's father. Can he do it?

On the one hand, it is easy to imagine Tim's specific act of killing his grandfather: he has the weapon and he can pull the trigger.

But were Tim to kill his own grandfather, his father would not have existed, and thus Tim would not have existed. Tim's grandfather did not in fact die, because Tim exists.

A future-oriented version of the Grandfather Paradox: *The Survival Paradox*.

Bad News: Sophie, who departs the year 2000 in her time machine and arrives in 2026, plans to live out her years in the future from 2026 onwards. Googling herself upon her arrival in 2026, she discovers a reliable news report of her death in a plane crash in 2010. Does Sophie have the ability to live out her years in 2026, or is she destined to return to the past in order to die in 2010?

Bootstrapping paradox: sometimes, things are responsible for their own existence due to time travel.

Example: you receive the plans for a time machine in the mail from yourself in the future. You build the time machine and travel to the future with it. Then you send the plans back in time to yourself. What started the loop?

A *causal loop* is a sequence of events in which each member is a cause of its successor, and where the final member is a cause of the first, but where there is no explanation for what actually begins the loop. The loop has no real origin.

Time travel in *Terminator 1*: John Connor sends Kyle Reese back in time to protect his mother Sarah Connor from a robot assassin, but Kyle Reese conceives John Connor with Sarah Connor in the past, and then dies in the past. John Connor's own existence depends on sending Kyle Reese back in time.

Secondary paradox: Kyle Reese was supposed to exist all the way until the time of grown-up John Connor in order to be sent back in time.

Time travel in *Terminator 2* involves a bootstrapping paradox as well. Skynet, the evil and all-encompassing AI, is based on technology sent back in time from the future. John Connor, Sarah Connor, and a non-evil Terminator destroy the technology upon which the future Skynet is based. But Skynet has to be in existence in the future in order to send the Terminator back which starts the chain of events in the movie.

A word about branching time travel: it's not always classified by philosophers as real time travel, since it technically involves travelling to, or bringing into being, different spatiotemporal structures than the actual world.

3. Dangerous things to Do While Time Travelling

- a. Remain located at the same spatial location for the duration of travel through time. (You will run into yourself or into something else on the preexisting timeline.)
- b. Stick your arm out the window of the time machine.
- c. Travel to a time that doesn't exist.

Philosophers disagree on the exact nature of time, and how much time exists at once. There are three prominent views:

Eternalism: past, present, and future are real.

Presentism: only the present is real.

Growing Block: only the past and the "bleeding edge" of the present are real.

- d. "Rewind" external time so that you don't know why you went back in time.
- e. Endanger your own existence by changing the past.

What to Expect on your Journey

1. Not much of a view from the window of the time machine.
2. Bilocation: you might meet an earlier version of yourself, so you will be two different ages at the same time.
3. A causal loop: something caused you to return to the past, which caused you to do something in the future to ensure that you would return to the past.

Fun Activities for Any Time Traveler

1. Invest \$1 so that it will grow to one million dollars by the time you are grown up.
2. Build a whole house out of a single time travelling brick from the future.
3. Set up your own biological parents so that they create you.

Puzzles and Questions

1. If you enter a time machine that takes one year to travel back in time, how much time has elapsed for people “on the ground” on the external timeline while you are travelling back in time?
2. If you travel into the past to meet a younger version of yourself, must you return to the future in order to remain logically consistent?
3. Does time travel to the future preclude the existence of free will?
4. Would the possibility of time travel carry any special moral obligations?

Thank you for coming! Email sara.bernstein@ucsc.edu if you have questions about time travel, metaphysics, the meaning of life, or the Philosophy Major at UCSC.